Understanding Invariance via Feedforward Inversion of Discriminatively Trained Classifiers

Piotr Teterwak, Chiyuan Zhang, Dilip Krishnan, Michael C. Mozer

Talk by Dmitry Vasilenko

Problem Statement and Motivation

Discriminative models should be invariant to irrelevant features, yet...

- Classifier: $\Phi : \mathcal{X} \to \mathbb{R}^C$
- Logits: $z = \Phi(x)$
- Invariance: $\Phi(T(x)) \approx \Phi(x)$ for class-preserving transformations T

Key Research Question

How much information about x is preserved in z?

Related Research: Feature Inversion Methods

Two Main Approaches for Representation Inversion

Optimization-Based Methods

- Gradient descent in image space
- Minimize: $L(x, x_0) = ||\Phi(x) - \Phi(x_0)||^2 + \lambda R(x)$

Drawbacks

- Sensitive to initialization
- Requires careful tuning of λ
- Slow iterative process

Learning-Based Methods

- Train decoder network on $\{$logits, pixels$\}$ pairs
- Feedforward inference after training

Evolution

- Initial: Pixel loss + image prior $\rightarrow$ blurry
- Improved: + Adversarial loss $\rightarrow$ better quality
- **Our work:** + Conditional projection + Conditional BN

Method: Architecture Overview

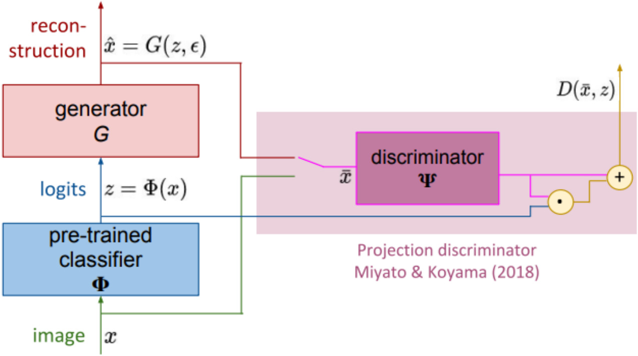


Figure: Architecture

GAN with Logit Conditioning

- Generator: $\hat{x} = G(z, \epsilon)$, $\epsilon \sim \mathcal{N}(0, I)$
- Discriminator: $D(\bar{x}, z)$

Mathematical Framework: Generator

Conditional Batch Normalization

$$y'_{l,k} = \gamma_{l,k}(z) \cdot \frac{y_{l,k} - \mathbb{E}[y_{l,k}]}{\sqrt{\mathbb{V}[y_{l,k}]}} + \beta_{l,k}(z)$$

Parameterization

- $\gamma_{l,k}(\cdot), \beta_{l,k}(\cdot)$ — MLPs mapping from z
- Logits control "style" at all abstraction levels

Mathematical Framework: Discriminator

Projection Discriminator

$$D(\bar{x}, z) = (w_1 + W_2 z)^\top \Psi(\bar{x})$$

Components

- $\Psi(\bar{x})$ — image feature vector
- $w_1^\top \Psi(\bar{x})$ — "realism" assessment
- $(W_2 z)^\top \Psi(\bar{x})$ — "consistency" assessment

Loss Function

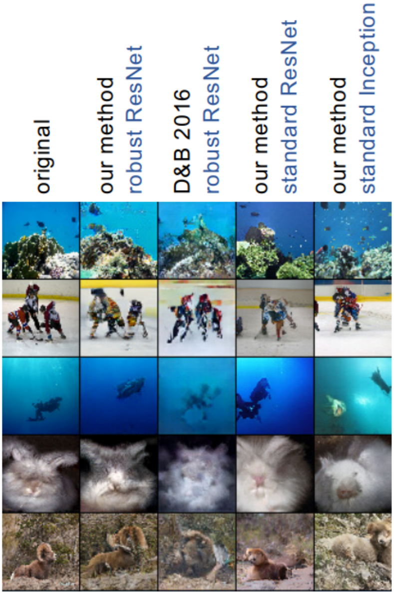
Minimax Objective

$$
\begin{aligned}
\mathcal{L}_D &= \mathbb{E}_{x,\epsilon}[\max(-1, D(G(z,\epsilon), z)) - \min(1, D(x, z))] \\
\mathcal{L}_G &= -\mathbb{E}_{x,\epsilon}[D(G(z,\epsilon), z)]
\end{aligned}
$$

Key Advantages

- Single loss term — no hyperparameter balancing
- Feedforward inference — fast reconstruction

Method Comparison



Human Evaluation Results

Comparison		
Ours (Robust ResNet) vs D&B Robust	87.5%	12.5%
Ours (Robust) vs Ours (Non-Robust)	76.5%	23.5%
Ours (ResNet) vs Ours (Inception)	52.7%	47.3%

Visualizing Model Invariances

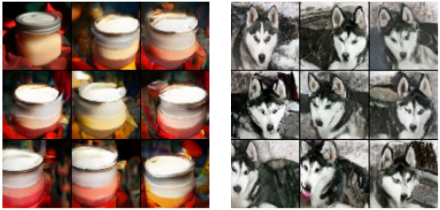
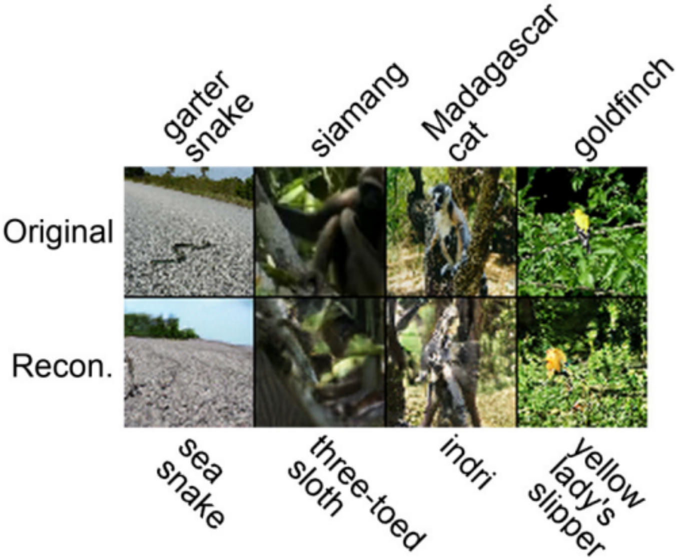


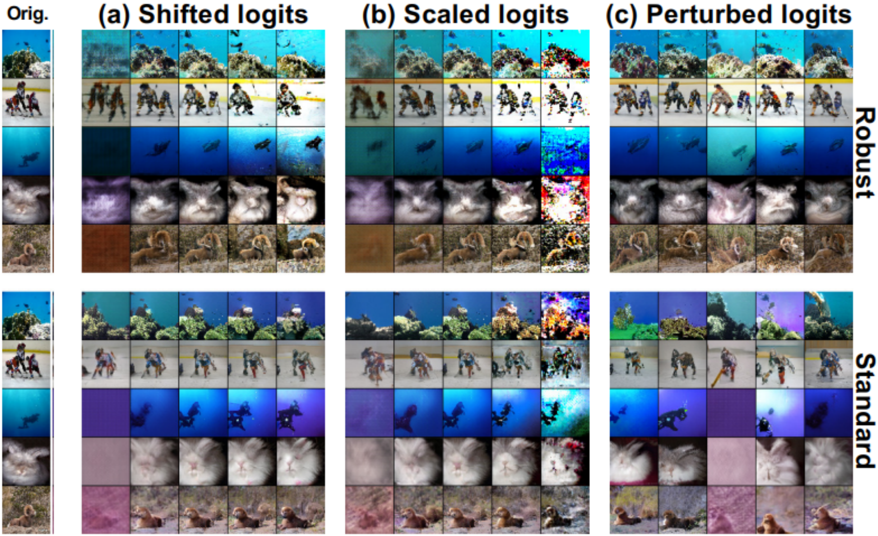
Figure: Robust ResNet-152

Figure: non-robust ResNet-152

Reconstructing incorrectly classified images



Logit manipulations



Conclusion

- We obtain remarkably high fidelity reconstructions, which allow the object class to be determined as well as visual detail orthogonal to the class label.
- Robust ResNet produces better reconstructions than non-robust ResNet
- Architecture matters. ResNet seems to better preserve visual information than Inception
- We do not see a qualitative difference in reconstruction fidelity for incorrectly classified images.
- For a robust ResNet-152, both logit shifts and rescaling have have a similar effect—they influence the contrast, sharpness, and brighness of reconstructions. The relationship for non-robust ResNet is similar but weaker